



Rafael Tiedra

Mathematical physicist with various years of experience in research, IT, education and administration.

Date of birth 05.03.1976

Nationality Swiss and Spanish

Languages French (native), English and Spanish (fluent)

Skills

Research and teaching

Problem-solving, research in physics and mathematics (40+ scientific publications)

Scientific presentation/vulgarization (150+ talks)

Classroom and online teaching

Academic and corporate administration

Management, budgets, fundraising, organisation of events and conferences

Writing and editing publications, popular science, translation, digital communication

Recruitment process, conception of curriculums, project management

Programming

Languages/skills: Python, SQL, TensorFlow, PyTorch, Scikit-Learn, VSCode, Jupyter, Git, OpenShift, Qiskit, CP-SAT, HTML, LaTeX

IT: MS Office, Linux, Windows OS

Professional experiences

- Aug. 2022-present **Data Scientist**, Department of Data Science and AI, Geneva State, Switzerland
Aug. 2021-July 2022 **Visiting professor**, Swiss Federal Institute of Technology Lausanne, Switzerland
Jan. 2009-Feb. 2022 **Professor**, Pontifical Catholic University, Chile
Oct. 2005-Dec. 2008 **Postdoctoral researcher**, Universities of Paris-Sud and Cergy-Pontoise, France

Studies

- June 2005 **PhD in Physics**, University of Geneva
Oct. 2001 **BSc in Mathematics**, University of Geneva
June 2001 **MSc in Physics**, University of Geneva

Prizes and research grants

- 2009, 2013, 2017, 2021 Chilean Fondecyt grants (501'000 USD)
2009, 2014 Chilean Milenio grants (2'338'000 USD)
2011 French-Chilean ECOS-CONICYT grant
2005, 2007 Swiss NSF grants (138'000 USD)
2001 Fanny Wurth scholarship
2001 Hans Wilsdorf scholarship (Rolex)
1998, 1999, 2000, 2001 Canton of Geneva scholarships